

WORKSHEET - NAPLAN Year 9

Number and Algebra Ratios, Scale and Proportion (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. Measurement and Geometry, NAP-08569

The distance between Newcastle and Canberra on the map below is 9 cm.



The scale of the map is 1 cm = 50 km.

What is the actual distance between Newcastle and Canberra?

350 km 450 km 600 km 900 km

☐ ☐ ☐ ☐

2. Measurement and Geometry, NAP-38007

Two places are 5.4 cm apart on a map.

On the map 1 cm represents 4 km.

What is the actual distance between the two places?

1.08 km 10.8 km 21.6 km 43.4 km

☐ ☐ ☐ ☐

3. Number and Algebra, NAP-02654

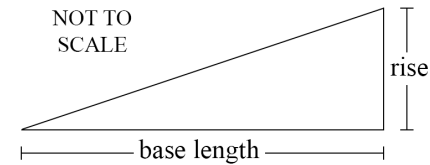
On a country property, 1 acre of land is recommended for every 4 sheep.

How many acres of land would be needed for 16 sheep?

4 6 16 64
☐ ☐ ☐ ☐

4. Measurement and Geometry, NAP-32121

A ramp rises 1 metre for every 8 metres of its base length.

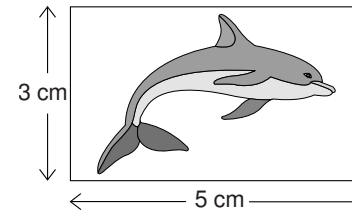


If it rises 0.7 metres, what is its base length?

5.6 metres 8.7 metres 11.4 metres 16 metres
☐ ☐ ☐ ☐

5. Number and Algebra, NAP-67473

Poppy uses a photocopier to enlarge this picture.



The enlarged picture is 3 times as high and 3 times as wide as the original.

The area of the enlarged picture is

☐ 3 times the area of the original.
☐ 6 times the area of the original.
☐ 9 times the area of the original.
☐ 24 times the area of the original.

6. Number and Algebra, NAP-02685

A christmas pudding recipe uses 3 cups of sugar for every 4 cups of sultanas.

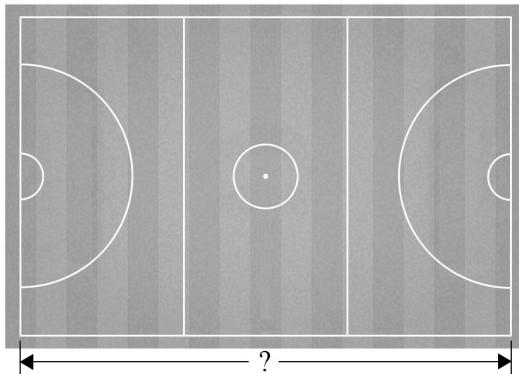
Select the correct combination of sugar and sultanas for this recipe.

- ☐ $\frac{1}{3}$ cup of sugar, $\frac{1}{4}$ cup of sultanas
- ☐ 1 cup of sugar, 1.5 cups of sultanas
- ☐ 1.5 cups of sugar, 2 cups of sultanas
- ☐ 4 cups of sugar, 5 cups of sultanas

7. Number and Algebra, NAP-85080

Libby plays on a hockey field that is 120 metres long.

She makes a scale diagram of the field using a ratio of 1 : 400.

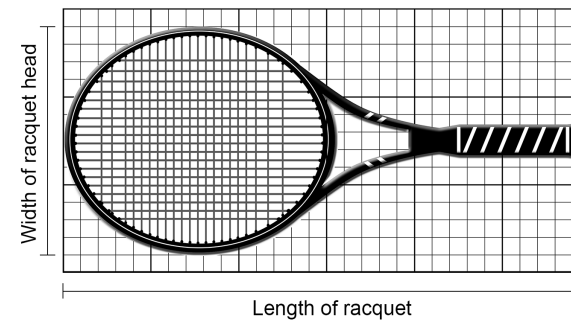


How long should Libby make the scale diagram of the field?

- 3 cm** **30 cm** **90 cm** **300 cm**
- ☐ ☐ ☐ ☐

8. Number and Algebra, NAP-32151

Jana drew a diagram of her tennis racquet.



The actual length of her racquet is 60 centimetres.

What is the actual width of her racquet head?

centimetres

Worked Solutions

1. Measurement and Geometry, NAP-08569

$$\begin{aligned}\text{Distance} &= 9 \times 50 \\ &= 450 \text{ km}\end{aligned}$$

2. Measurement and Geometry, NAP-38007

$$\begin{aligned}\text{Actual distance} &= 5.4 \times 4 \\ &= 21.6 \text{ km}\end{aligned}$$

3. Number and Algebra, NAP-02654

$$\begin{aligned}\text{Acres} &= \frac{16}{4} \\ &= 4\end{aligned}$$

4. Measurement and Geometry, NAP-32121

$$\begin{aligned}\text{Base length} &= 8 \times 0.7 \\ &= 5.6 \text{ metres}\end{aligned}$$

5. Number and Algebra, NAP-67473

Area of original picture

$$\begin{aligned}&= 3 \times 5 \\ &= 15 \text{ cm}^2\end{aligned}$$

Area of enlarged picture

$$\begin{aligned}&= 9 \times 15 \\ &= 135 \text{ cm}^2\end{aligned}$$

Times larger vs original

$$\begin{aligned}&= \frac{135}{15} \\ &= 9 \text{ times}\end{aligned}$$

6. Number and Algebra, NAP-02685

$$\begin{aligned}\text{Ratio} &= 3:4 \\ &= 1.5:2\end{aligned}$$

$\therefore$ 1.5 cups of sugar, 2 cups of sultanas is
in the correct ratio.

7. Number and Algebra, NAP-85080

$$\begin{aligned}\text{Ratio is } &1:400 \\ \text{Since } 120 \text{ m} &= 12\,000 \text{ cm,} \\ \text{Length of the scale diagram} \\ &= 12\,000 \div 400 \\ &= 30 \text{ cm}\end{aligned}$$

8. Number and Algebra, NAP-32151

Racquet is 30 grid intervals long.

$$\begin{aligned}30 \text{ intervals} &= 60 \text{ cm} \\ \therefore 1 \text{ grid interval} &= 2 \text{ cm}\end{aligned}$$

Racquet head is 13 grid intervals high,

$$\begin{aligned}\therefore \text{Head width} &= 13 \times 2 \\ &= 26 \text{ cm}\end{aligned}$$